ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** lie in the same plane	All three are coplanar.
2	**(d)** diffuse reflection	Irregular.
3	**(b)** R = 2f	Focus is midway.
4	**(d)** angle of incidence equals angle of reflection	i = r.
5	**(a)** regular reflection	Parallel beam stays parallel.
6	**(c)** convex	Wide field of view.
7	**(c)** 35 degrees	i = r.
8	**(a)** reflects light into our eyes	Reflection.
9	**(b)** concave mirror	Magnified upright image.
10	**(c)** normal	Always from the normal.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Incident ray, reflected ray and normal lie in the same plane (1); Angle of incidence equals angle of reflection (1).	1+1
12	Left appears as right in the mirror (1); So drivers can read it correctly in the rear-view mirror (1).	1+1
13	Convex lens is thick in the middle and converges light (1); Concave lens is thin in the middle and diverges light (1).	1+1
14	Regular: from smooth surface, parallel beam stays parallel (1); Diffuse: from rough surface, scattered (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Incident ray strikes mirror at the point O (1); Normal drawn at O (1); Reflected ray makes angle r = i on the other side (1).	1+1+1
16	Reflecting surface curves inwards vs outwards (1); Concave converges rays; convex diverges (1); Concave can form real images; convex only virtual (1).	1+1+1
17	Spectacles (1); Cameras (1); Magnifying glass or microscope (1).	1+1+1
	OR: Angle with the normal = 90 - 40 (1); i = 50 degrees (1); r = 50 degrees (1).	1+1+1
18	Pole: centre of the mirror surface (1); Principal axis: line through C and P (1); Focus: where parallel rays meet or appear to meet (1).	1+1+1
	OR: Parallel rays converge to the focus (1); A bright spot forms on the screen (1); Image is real and inverted (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Incident and reflected ray on same side of normal (1); i = r (1); Coplanar (1); Activity: pins and protractor on paper (1); Measured angles are equal (1).	1+1+1+1+1
	OR: Convex converges (1); Concave diverges (1); Focus and focal length (1); Convex: magnifier, camera (1); Concave: spectacles for short sight (1).	1+1+1+1+1
20	Virtual, erect, same size (1); Laterally inverted (1); Image distance = object distance (1); n = 360/90 - 1 = 3 (1); Images formed by multiple reflections (1).	1+1+1+1+1
	OR: Hold the lens facing the Sun (1); Place a screen behind it (1); Adjust until a bright spot forms (1); Measure the distance (1); It is the focal length (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 3	n = 360/90 - 1.
2	**(d)** concave mirror	Enlarged image.
3	**(a)** 0 degrees	i = 0, so r = 0.
4	**(a)** multiple reflections	Mirrors at angle.
5	**(b)** centre	Bulges in the middle.
6	**(d)** virtual and erect	Cannot be obtained on a screen.
7	**(b)** two plane mirrors	To see over obstacles.
8	**(d)** converging lens	Converges.
9	**(c)** virtual image	Virtual.
10	**(b)** concave mirror	Like a cave.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Focal length: distance from the pole to the focus (1); Radius of curvature: radius of the sphere of which the mirror is a part (1).	1+1
12	A lens used to see small objects enlarged (1); Convex lens (1).	1+1
13	Virtual and erect (1); Same size, laterally inverted, as far behind as object is in front (1).	1+1
14	Shaving mirrors (1); Torch, headlight reflectors, dentists' mirrors (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Wall surface is rough (1); Reflection is diffuse (1); No definite image is formed (1).	1+1+1
16	f = R/2 (1); = 30/2 (1); = 15 cm (1).	1+1+1
17	It converges the Sun's rays to a point (1); Heat energy concentrates (1); Paper ignites (1).	1+1+1
	OR: Virtual and erect (1); Same size (1); Image distance equals object distance (1).	1+1+1
18	They give an erect image (1); Always diminished images give a wide field of view (1); Driver sees more traffic (1).	1+1+1
	OR: It diverges the rays (1); Rays appear to come from a point (1); The image is virtual (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Three long mirror strips at 60 degrees (1); Placed in a tube with coloured beads (1); Multiple reflections give a pattern (1); Rotating changes the pattern (1); Number of images depends on angle (1).	1+1+1+1+1
	OR: Eye lens forms real image on retina (1); Myopia: image falls in front of retina (1); Concave lens diverges rays to correct (1); Hypermetropia: image behind retina (1); Convex lens converges rays (1).	1+1+1+1+1
20	Concave and convex (1); Pole, principal axis (1); Centre of curvature (1); Principal focus (1); f = R/2 (1).	1+1+1+1+1
	OR: Reflected rays diverge (1); They appear to come from a point behind the mirror (1); Image distance equals object distance (1); Virtual (1); Same size (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** diverging lens	Diverges.
2	**(d)** of the same size as the object	Same size.
3	**(b)** convex mirror	Reflecting surface bulges out.
4	**(c)** convex lens	Magnifies.
5	**(d)** 4 m	Image is 2 m behind.
6	**(d)** principal focus	Focus.
7	**(a)** concave lenses	Diverge rays.
8	**(b)** laterally inverted	Left-right reversed.
9	**(c)** focal length	f.
10	**(d)** virtual and diminished	Always smaller.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Image is 10 cm behind the mirror (1); Distance between object and image = 20 cm (1).	1+1
12	Image that cannot be obtained on a screen (1); Image in a plane mirror (1).	1+1
13	Rear-view mirrors (1); Security and street mirrors (1).	1+1
14	Kaleidoscope (1); Periscope (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	n = 360/60 - 1 (1); = 6 - 1 (1); = 5 images (1).	1+1+1
16	Bulb is placed at the focus (1); Reflected rays form a parallel beam (1); Light goes to long distances (1).	1+1+1
17	Image is formed behind the mirror, flipped along the depth direction (1); Appears laterally inverted (1); Up stays up (1).	1+1+1
	OR: Two plane mirrors parallel at 45 degrees to the tube (1); Light reflects twice (1); Allows viewing over an obstacle (1).	1+1+1
18	Shape: thick vs thin in the middle (1); Converges vs diverges (1); Real or virtual vs virtual image (1).	1+1+1
	OR: Real image forms on a screen; virtual does not (1); Real is inverted; virtual is erect (1); Real formed by actual rays meeting; virtual by apparent meeting (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Held close: virtual, erect and enlarged image (1); Held far: real and inverted image (1); The enlarged image helps to see details (1); Dentist: enlarged view of the teeth (1); Barber: enlarged view of the face in shaving (1).	1+1+1+1+1
	OR: Periscope: plane mirrors (1); Camera: convex lens (1); Microscope: lenses (1); Telescope: lenses or mirrors (1); Image formation is basic (1).	1+1+1+1+1
20	Plane: dressing, periscope (1); Concave: shaving, dentist (1); Concave: torch reflector (1); Convex: rear-view (1); Convex: street mirror (1).	1+1+1+1+1
	OR: Smooth surface: regular reflection (1); Rough surface: diffuse reflection (1); Light scattered in all directions lets us see objects (1); Mirror gives definite image (1); Both obey laws of reflection (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.